Mzuzu University

Department of Mathematics and Statistics

MATH 1101 - Precalculus

Supplementary/Deferred Examination

Friday, December 17, 2021 Time Allowed: 3 hours (2:00 - 5:00 pm)

INSTRUCTIONS

1. **Calculators** are **not allowed**.
2. Write the name of lecturer on the cover page of your answer book.
3. Check that the paper has **11** questions.
4. Attempt **all** questions.
5. Your **answers should be presented in order** of the questions.
6. Show **all** your working and use appropriate notation.

1. (a) Simplify $\left(\frac{q^{-1}r^{-1}s^{-2}}{r^{-5}sq^{-8}}\right)^{-1}$. [3]

   (b) Simplify $x(x-y)^{-1} - y(y-x)^{-1}$. [3]

   (c) Factorise $8a^3 + 27b^3 + 2a + 3b$. [3]

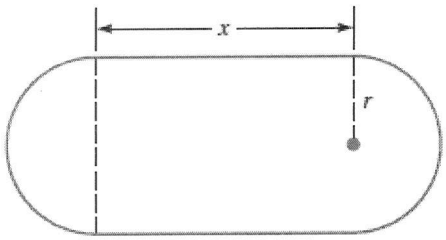
2. (a) Determine whether the equation $y = \frac{1}{-x^2}$ is symmetrical about the x-axis, y-axis or the origin. [3]

   (b) Find the point with coordinates of the form $(2a, a)$ that is in third quadrant and is at a distance of 5 from $P(1, 3)$. [3]

3. (a) Let $g(x) = 4x - 1$. Find $f(x)$, given that the equation $g(f(x)) = x + 5$ is true for all values of x. [3]

   (b) Let $f(x) = 4x^2$. Evaluate $\frac{f(x)-f(-2)}{x-(-2)}$. [4]

4. An athletic field with a perimeter of $\frac{1}{4}$ mile consists of a rectangle with a semicircle at each end, as shown in the figure below. [6]



Find the dimensions r that yield the greatest possible area for the rectangular region.

5. (a) Let $S(x) = \dfrac{2^x - 2^{-x}}{2}$ and $C(x) = \dfrac{2^x + 2^{-x}}{2}$. Find $(C(x))^2 - (S(x))^2$. [4]

   (b) Let $f(x) = \ln\left(1 - \dfrac{1}{x^2}\right)$. Find $f(2) + f(3) + f(4)$. [3]

6. Solve for x in each of the following equations:

   (a) $5\sqrt[3]{x^4} - 13\sqrt[3]{x^2} + 8 = 0$, [6]

   (b) $\dfrac{\ln(\sqrt{x+4}+2)}{\ln\sqrt{x}} = 2.$ [5]

7. Find the solution set for each of the following inequalities and present your answer using interval notation:

   (a) $|3(x+2)^2 - 3x^2| < k$ where $k > 0$. [5]

   (b) $\log_2 x - \log_2(x+1) - \log_2(2x+6) < 0$. [7]

8. Sketch the graph of $y = e^{-x} - e$ and find the domain, range, x-and y-intercepts, and asymptotes. [5]

9. Convert

   (a) $-540°$ to radians, [2]

   (b) $\frac{13\pi}{3}$ radians to degrees. [2]

10. Evaluate

    (a) $\sin(-945°)$ [4]

    (b) $\csc\beta$ if $\sec\beta = -\frac{3}{2}$ and $180° < \beta < 270°$. [4]

    (c) $\ln(\sin^2 17° + \sin^2 73°) - \cos(\ln e^{180°})$ [4]

11. (a) Show that $\dfrac{2\sin^3\beta}{1-\cos\beta} = 2\sin\beta + 2\sin\beta\cos\beta.$ [4]

    (b) In the expression $\sqrt{9-x^2}$, make the substitution $x = 3\sin\theta$ for $0 < \theta < \frac{\pi}{2}$, and show that the result is $3\cos\theta$. [3]